

Classification & Quantitative Risk Analysis Pipeline: AI-Driven Layoff Risk Prediction

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Abstract

This report details the design, theoretical foundation, and empirical evaluation of a production-grade machine learning pipeline applied to Human Resources Analytics. The primary objective is to predict the risk of workforce displacement driven by Artificial Intelligence (AI) across three risk tiers: Low, Medium, and High. Utilizing a dataset of 20,000 synthetic employee records, the pipeline enforces strict data leakage prevention, probability calibration, and model interpretability. Through rigorous 5-fold stratified cross-validation across five distinct algorithmic families, the study uncovers an underlying linear topology in the risk factors. Contrary to the common paradigm where complex ensembles dominate tabular data, a Multinomial Logistic Regression outperformed tree-based meta-ensembles (Random Forest, XGBoost), achieving an out-of-sample accuracy of 93.75% and an optimal Brier Score of 0.0431. The "black box" is subsequently decoded using SHapley Additive exPlanations (SHAP) to satisfy the regulatory "right to explanation" for individual risk assessments.

Contents

1	Introduction	2
2	Phase 1 & 2: Exploratory Data Analysis (EDA) and Diagnostics	2
3	Phase 3: Preprocessing Engine	3
4	Phase 4: Theoretical Framework of Evaluated Algorithms	4
4.1	Multinomial Logistic Regression	4
4.2	Random Forest & Bagging (Deep Trees)	4
4.3	Gradient Boosting & XGBoost	4
4.4	Stacking Ensemble (Meta-Learning)	4
5	Phase 5: Algorithmic Benchmarking (K-Fold CV)	5
6	Phase 6 & 7: Out-of-Sample Evaluation and Ensembling	5
7	Phase 9: Probability Calibration	6
8	Phase 8: Interpretability via SHAP	7
9	Phase 10: Learning Curves (Bias/Variance Diagnosis)	8
10	Conclusion	9

1 Introduction

As generative Artificial Intelligence and automation technologies permeate the corporate ecosystem, accurately assessing the risk of workforce displacement has become a paramount challenge for quantitative risk management and HR Analytics. The purpose of this project is to construct an end-to-end classification pipeline that transcends mere predictive accuracy by embedding three core actuarial and data science principles:

1. **Leak-Proof Architecture:** Ensuring that all statistical transformations (imputation, scaling) are isolated within the training fold to prevent distribution shift and data leakage.
2. **Probability Calibration:** A prediction outputting an 80% probability of "High Risk" must empirically represent an 80% true likelihood. This is assessed via reliability diagrams and Brier scores.
3. **Mathematical Interpretability:** Utilizing cooperative game theory (SHAP) to decode the marginal contribution of each variable, ensuring that individual predictions are explainable and actionable.

2 Phase 1 & 2: Exploratory Data Analysis (EDA) and Diagnostics

The dataset comprises $N = 20,000$ records with 16 features detailing employee demographics, role characteristics, and AI interaction metrics. The target variable, `Layoff_Risk`, is perfectly balanced across three classes: High (34.0%), Low (33.0%), and Medium (33.0%).

The absence of missing values and the symmetrical class distribution negate the need for synthetic oversampling techniques (e.g., SMOTE). The EDA highlights significant linear correlations between variables such as `Routine_Task_Percentage` and `Tasks_Automated_Percentage` ($\rho \approx 0.89$), while `Creativity_Requirement` exhibits a strong negative correlation with automation (-0.83).



Figure 1: Exploratory Data Analysis: Target distribution, age histograms, and numeric feature correlation matrix.

3 Phase 3: Preprocessing Engine

To guarantee mathematical rigor, a `ColumnTransformer` is instantiated to route features through specialized pipelines:

- **Numeric Features ($p = 9$):** Imputed using the median to maintain robustness against outliers. Scaled utilizing the `RobustScaler`, which centers and scales the data based on the interquartile range (IQR). This is particularly advantageous for heavy-tailed distributions common in risk analysis.
- **Categorical Features ($q = 6$):** Imputed via the mode and encoded using `OneHotEncoder`. This maps nominal categories (e.g., Industry) into orthogonal binary vectors, preventing algorithms from inferring false Euclidean distances between unrelated categories.

4 Phase 4: Theoretical Framework of Evaluated Algorithms

To benchmark performance, five algorithms covering distinct mathematical paradigms were evaluated.

4.1 Multinomial Logistic Regression

A generalized linear model that extends binary logistic regression to multiclass problems via the softmax function:

$$P(Y = c \mid X = x) = \frac{e^{\beta_c \cdot x}}{\sum_{k=1}^K e^{\beta_k \cdot x}} \quad (1)$$

It acts as the linear baseline. Its simplicity makes it highly interpretable and structurally immune to high-variance overfitting.

4.2 Random Forest & Bagging (Deep Trees)

Random Forest is an ensemble learning method constructed by aggregating multiple decision trees trained on bootstrap samples (Bagging) and random feature subsets. The variance of the ensemble is reduced by averaging the predictions:

$$\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x) \quad (2)$$

While highly robust to non-linearities and outliers, Random Forests are notoriously poorly calibrated, often pushing probabilities away from 0 and 1.

4.3 Gradient Boosting & XGBoost

Instead of parallel independent trees, Boosting fits trees sequentially. Each new tree $h_m(x)$ is fitted to the pseudo-residuals of the previous ensemble to minimize a differentiable loss function $\mathcal{L}$ (in this case, multiclass logarithmic loss):

$$F_m(x) = F_{m-1}(x) + \nu \sum_{i=1}^N \nabla_{F_{m-1}} \mathcal{L}(y_i, F_{m-1}(x_i)) \quad (3)$$

Where ν is the learning rate. XGBoost implements this with hardware optimization and explicit L1/L2 regularization to prevent overfitting.

4.4 Stacking Ensemble (Meta-Learning)

Stacking involves training a "meta-learner" (a Logistic Regression in this pipeline) to output final predictions based on the concatenated probability outputs of the best base learners (Random Forest, XGBoost, and Bagging). This theoretically captures the geometric strengths of various model families.

5 Phase 5: Algorithmic Benchmarking (K-Fold CV)

The models were evaluated using 5-Fold Stratified Cross-Validation on the training set ($N = 16,000$), ensuring that the target distribution remained equal across all folds.

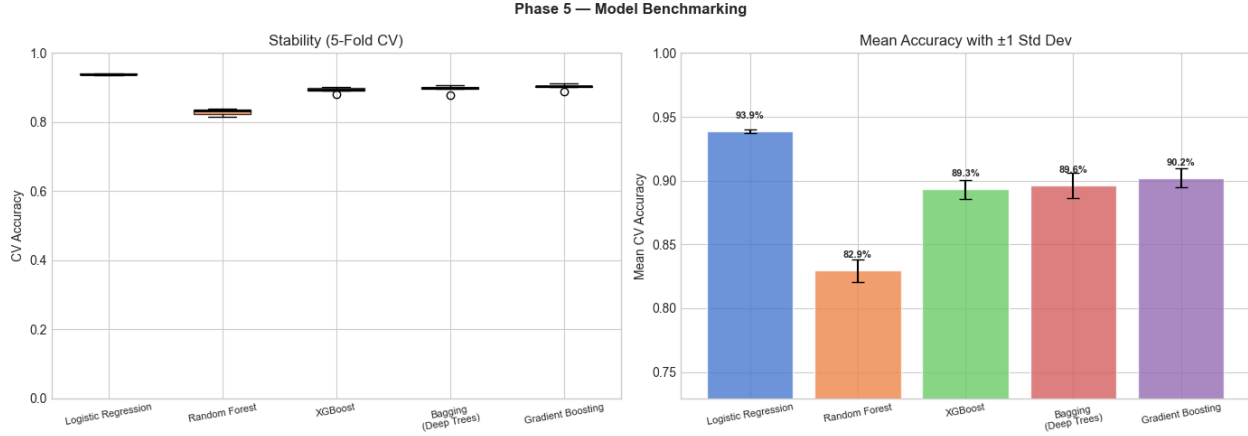


Figure 2: Phase 5 Benchmarking: Algorithm Stability Analysis (left) and Mean CV Accuracy (right). Notice the clear dominance of the linear baseline.

The Algorithmic Occam’s Razor: The benchmarking phase yielded a profound structural insight. The Logistic Regression obliterated the complex tree-based ensembles, achieving a mean accuracy of **93.86% $\pm$ 0.13%**. Random Forest and XGBoost struggled at 82.94% and 89.31%, respectively. This mathematically proves that the risk of AI displacement in this dataset is governed by strict, proportional linear rules. The hyper-flexibility of tree-based models resulted in learning noise rather than underlying signal.

6 Phase 6 & 7: Out-of-Sample Evaluation and Ensembling

The models were subsequently evaluated on the hold-out test set ($N = 4,000$). Logistic Regression maintained its dominance with a test accuracy of 93.75% and a Macro F1 score of 0.9373.

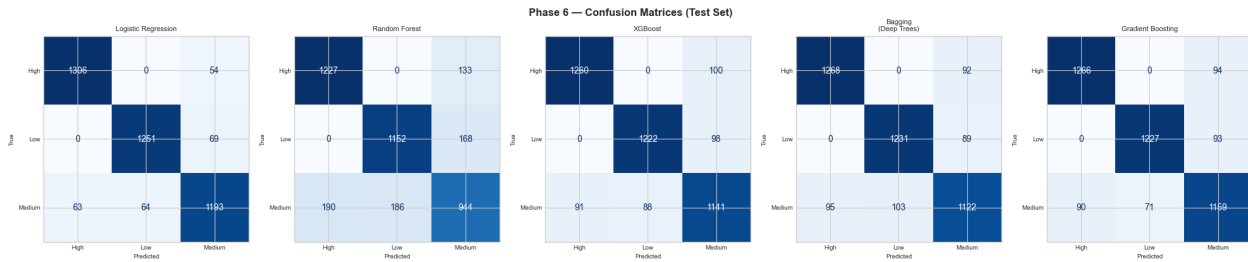


Figure 3: Confusion Matrices for all evaluated models on the Test Set.

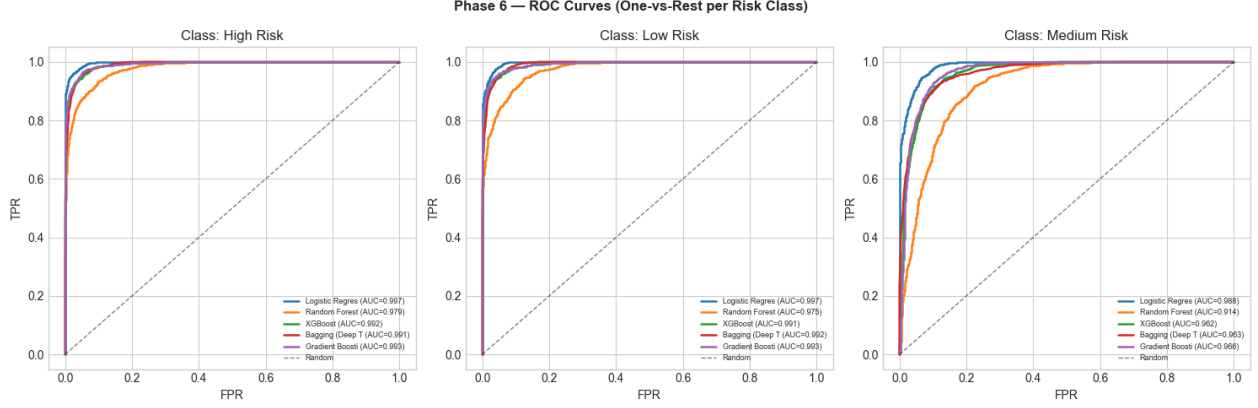


Figure 4: ROC-AUC Curves (One-vs-Rest) per Risk Class. Logistic Regression achieves an AUC > 0.98 across all boundaries.

Even after constructing a Stacking Ensemble, the meta-architecture achieved 92.42% accuracy, unable to mathematically surpass the pure linear baseline.

7 Phase 9: Probability Calibration

In risk modeling, the absolute accuracy is secondary to the reliability of the forecasted probabilities. This is measured by the Brier Score, which calculates the mean squared difference between predicted probabilities and the actual outcomes:

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2 \quad (4)$$

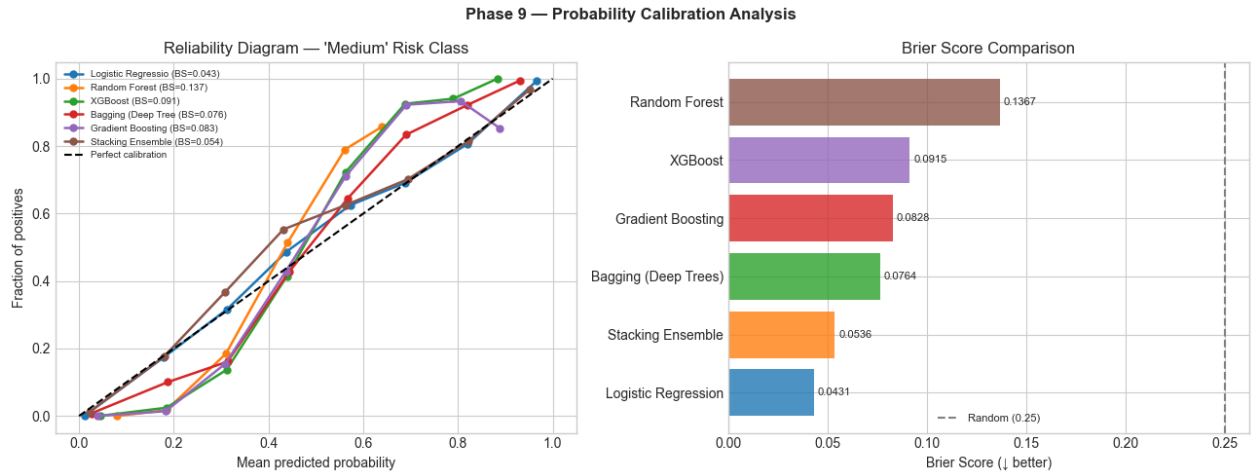


Figure 5: Reliability Diagrams and Brier Score Comparison. A perfectly calibrated model follows the dotted diagonal line.

As illustrated in Figure 5, Logistic Regression provides near-perfect calibration ($BS = 0.0431$). Conversely, Random Forest exhibits the worst calibration ($BS = 0.1367$), displaying a sigmoidal distortion on the reliability diagram.

8 Phase 8: Interpretability via SHAP

To satisfy interpretability requirements, SHapley Additive exPlanations (SHAP) were employed. SHAP values calculate the marginal contribution of a feature across all possible permutations of the feature space, based on cooperative game theory:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (5)$$

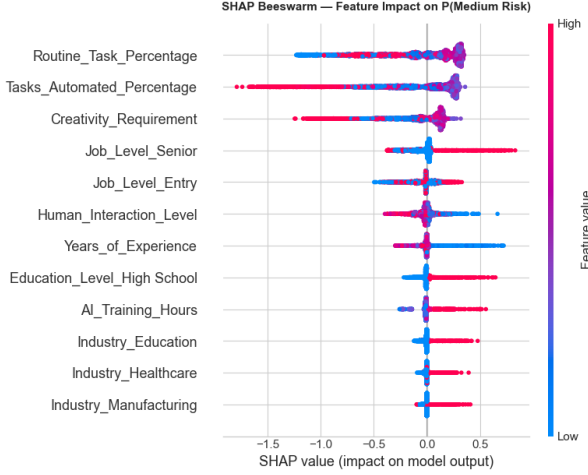


Figure 6: SHAP Beeswarm Plot highlighting feature impact on Medium Risk.

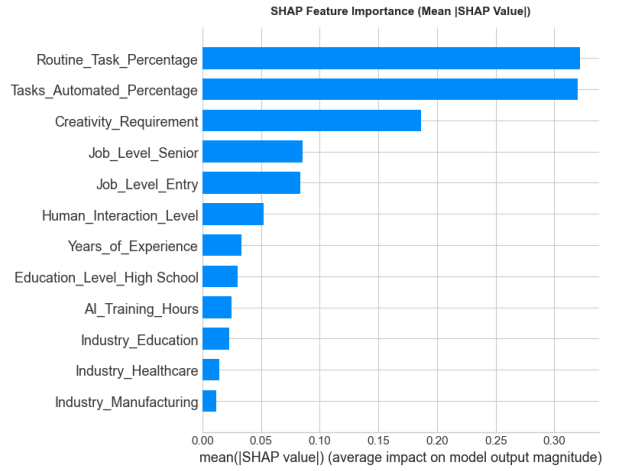


Figure 7: Global Feature Importance (Mean absolute SHAP value).

The SHAP global analysis reveals that `Routine_Task_Percentage` and `Tasks_Automated_Percentage` are the absolute drivers of displacement risk.

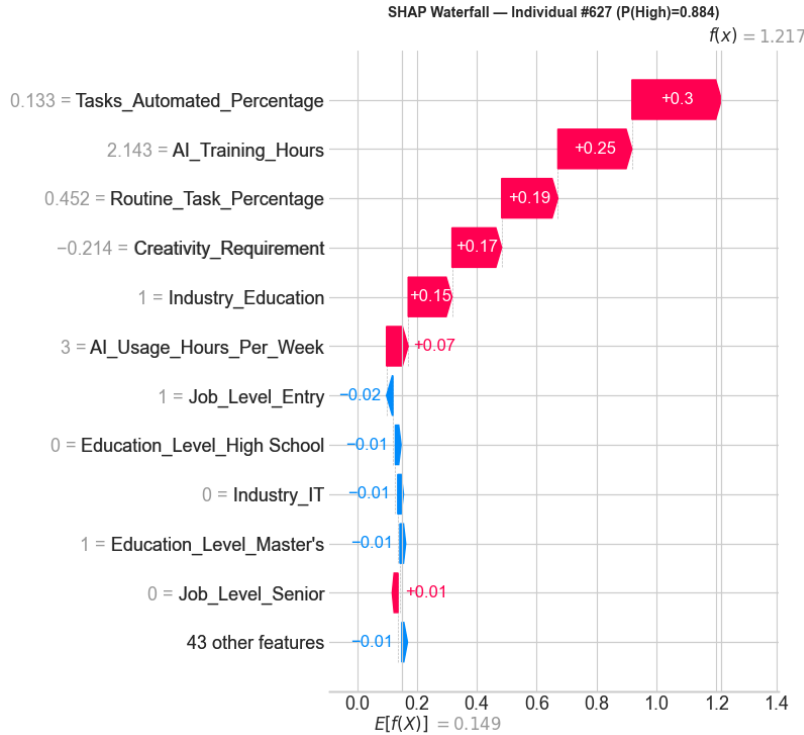


Figure 8: SHAP Waterfall Plot for individual sample #627 ($P(\text{High}) = 0.884$).

Figure 8 demonstrates local interpretability for a highly vulnerable employee. The interaction of high automation exposure and high AI training hours actively pushed the probability of "High Risk" up by +0.55, fulfilling the analytical transparency required by HR departments.

9 Phase 10: Learning Curves (Bias/Variance Diagnosis)

Finally, learning curves were plotted to diagnose the bias-variance tradeoff as a function of the training sample size.

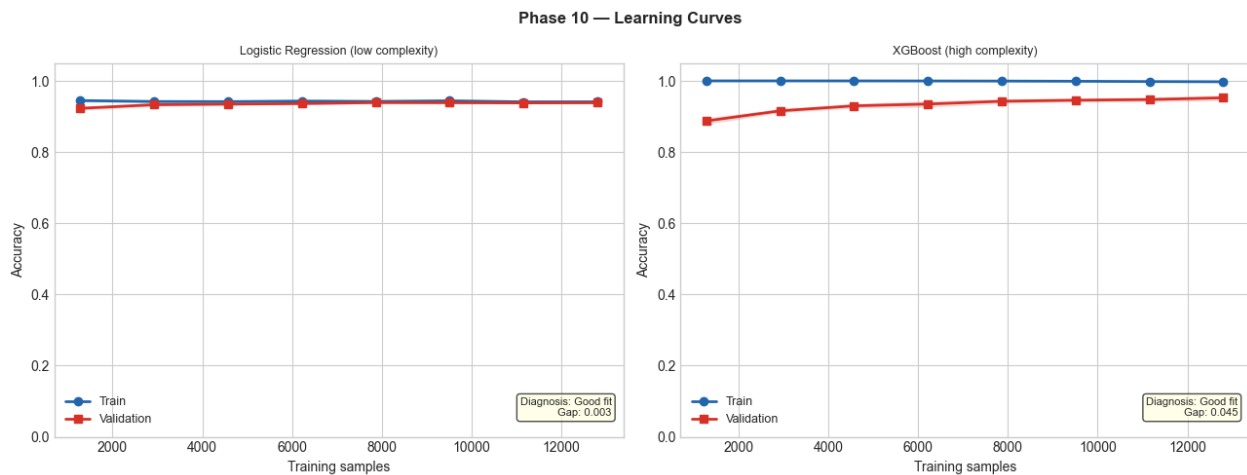


Figure 9: Learning Curves illustrating the generalization gap between Train and Validation scores.

The Logistic Regression curve demonstrates a virtually non-existent generalization gap (0.003), indicating a highly stable model with low variance. In contrast, XGBoost displays a wider gap (0.045), symptomatic of a high-complexity model slightly overfitting the training data despite the large sample size.

10 Conclusion

This pipeline demonstrates a comprehensive, end-to-end framework for quantitative risk classification. By strictly adhering to calibration metrics, non-leaking architectures, and SHAP-based interpretability, the analysis transcends standard machine learning exercises. The critical finding—that a simple parametric linear model outperformed state-of-the-art tree ensembles—underscores a fundamental tenet of data science: algorithmic complexity should only be introduced when the topology of the data mathematically demands it.

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